Euler's Method and Logistic Growth

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

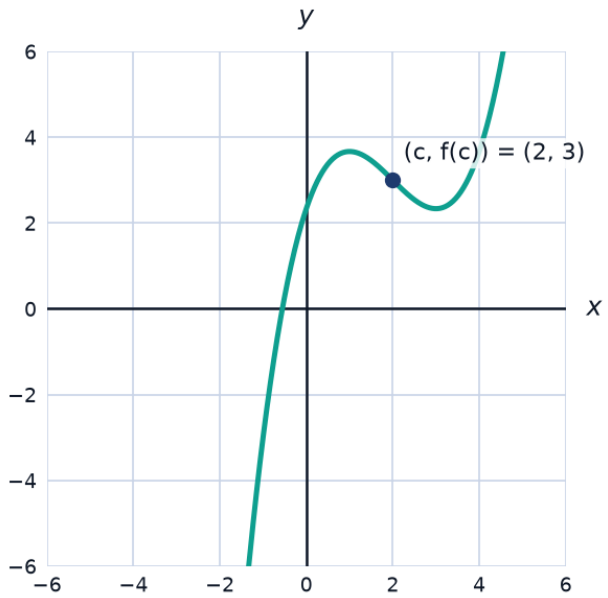
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Euler's Method Iterations

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

2 [Foundational] • Target: The Logistic Differential Equation

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.



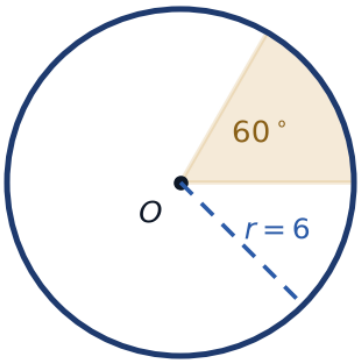
3 [Foundational] • Target: Analytical Solutions to Logistic Models

A particle's position in the xy-plane is given by the parametric equations $x(t) = at^2 - 3t$ and $y(t) = 2t^3 + bt$, where a and b are constants. At time $t = 1$, the particle's velocity vector is $\mathbf{v}(1) = \langle 9, 10 \rangle$. Find the value of a and the value of b.

© ClipSAT — clipsat.org | All Rights Reserved

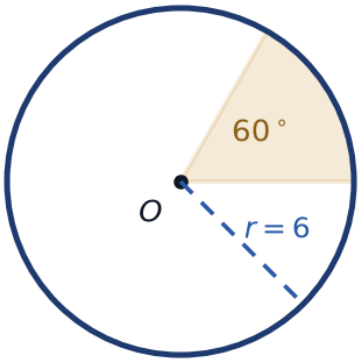
4 [Foundational] • Target: Advanced Slope Fields with Autonomous Equations

Evaluate $\lim_{x \to 0} \frac{\tan x}{x}$.



5 [Foundational] • Target: Euler's Method Iterations

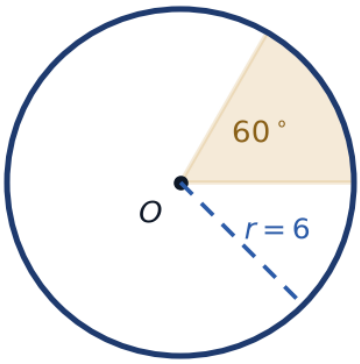
For $x = \cos t,\ y = \sin t$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{2}$.



© ClipSAT — clipsat.org | All Rights Reserved

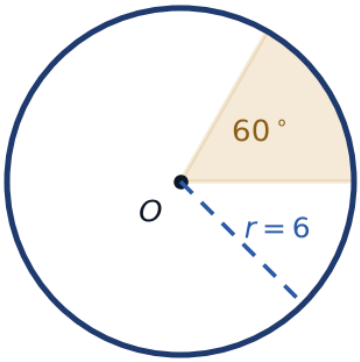
6 [Foundational] • Target: The Logistic Differential Equation

Find the area enclosed by the polar curve $r = 2$.



7 [Foundational] • Target: Analytical Solutions to Logistic Models

Give the first three nonzero terms of the Maclaurin series for $\sin x$.

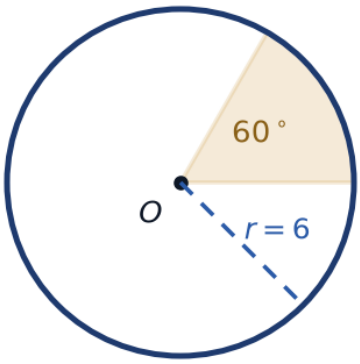


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

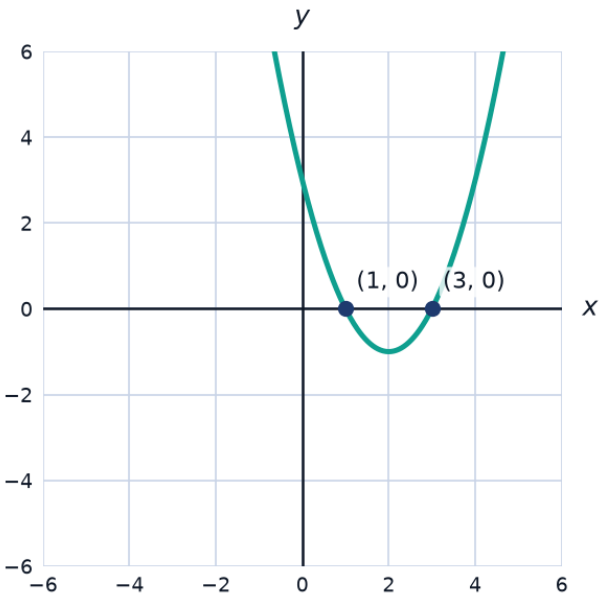
8 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

Give the first three terms of the Maclaurin series for $\cos x$.



9 [Intermediate] • Target: Euler's Method Iterations

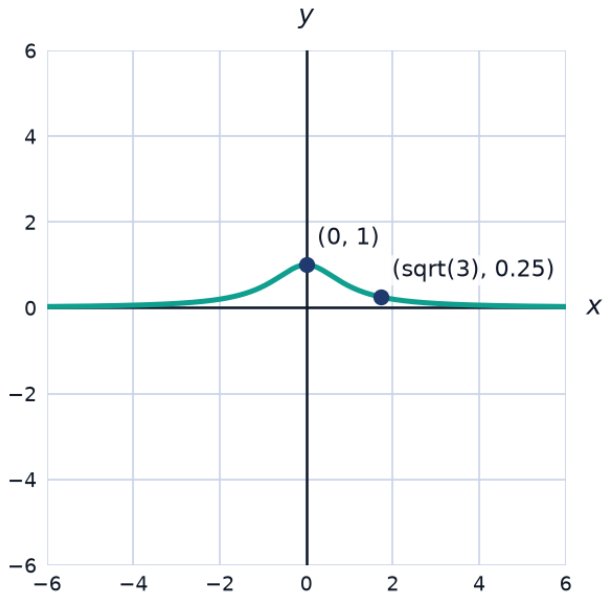
The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the definite integral of $f(x)$ from $x = 1$ to $x = 3$?



© ClipSAT — clipsat.org | All Rights Reserved

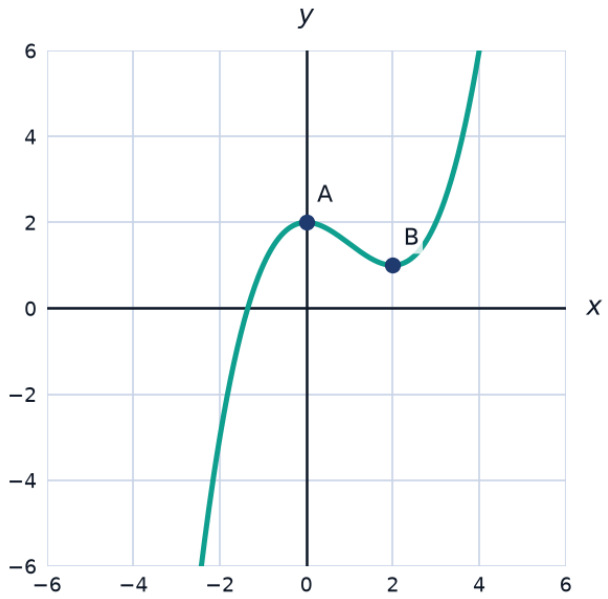
10 [Intermediate] • Target: The Logistic Differential Equation

Consider the region bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?



11 [Intermediate] • Target: Analytical Solutions to Logistic Models

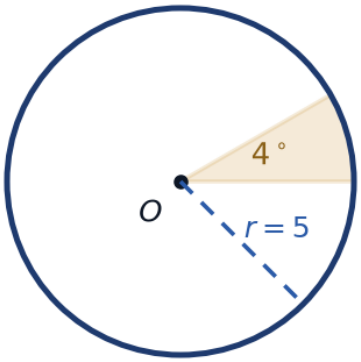
Evaluate $\int_0^2 6x^2\,dx$.



© ClipSAT — clipsat.org | All Rights Reserved

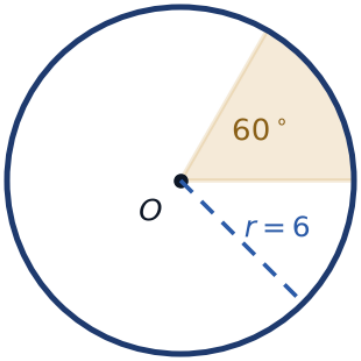
12 **[Intermediate]** • **Target: Advanced Slope Fields with Autonomous Equations**

A point is represented in polar coordinates by $(r, \theta) = (5, \frac{\pi}{4})$. What are the equivalent Cartesian coordinates (x, y)?



13 **[Intermediate]** • **Target: Euler's Method Iterations**

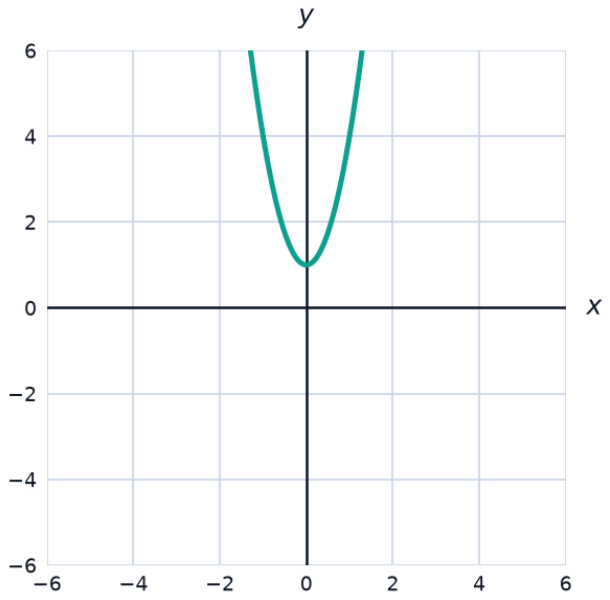
Evaluate $\int x \sin x \, dx$.



© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: The Logistic Differential Equation

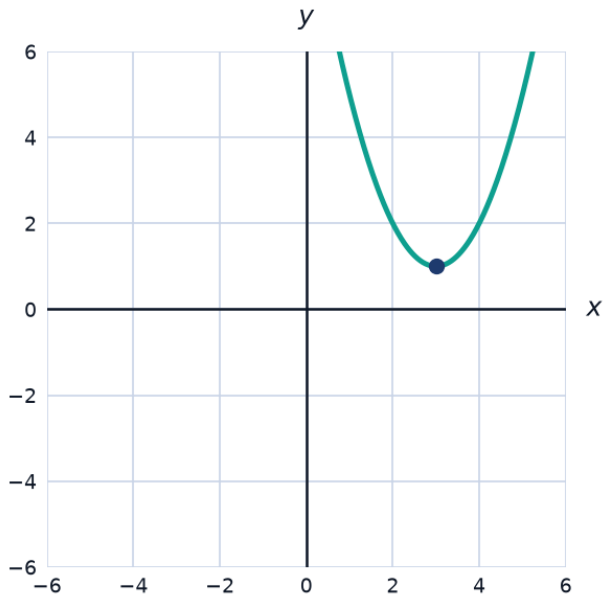
A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.



© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For what x does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?

17 [Challenging] • Target: Euler's Method Iterations • Cross-Lesson: 6.01 Power Series and Interval of Convergence

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

18 [Challenging] • Target: The Logistic Differential Equation • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the area enclosed by the cardioid $r = 1 + \cos\theta$.

19 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 1.01 Integration by Parts

Consider the function $f(x)$ defined by its Maclaurin series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!\, 3^{2n+1}}$. Let $g(x) = \int_0^x f(t)dt$. The function $f(x)$ is a known elementary function. Determine the value of $g''(0)$.

20 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Evaluate $\int_1^{\infty} \frac{1}{x^2}\, dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Euler's Method Iterations

Given $\frac{dy}{dx} = x + y$ and initial condition $y(0) = 1$, use Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$.

a. 2.25

b. 2.00

c. 2.50

d. 2.75

22 [Foundational] • Target: Euler's Method Iterations

If $y'' > 0$ (concave up) on the interval, does Euler's method yield an overestimate or underestimate of the true solution?

a. An underestimate because tangent segments lie below the curve

b. An overestimate because tangent segments lie above the curve

c. Exact value

d. Depends on the step size

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Euler's Method Iterations

Given $\frac{dy}{dx} = 2x$ with $y(1) = 3$, what is the error in approximating $y(2)$ using Euler's method with $\Delta x = 0.5$?

a. 0.5

b. 1.0

c. 0

d. 0.25

24 [Foundational] • Target: The Logistic Differential Equation

A population grows according to $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$. What is the carrying capacity of this population?

a. 500

b. 0.04

c. 250

d. $12,500$

25 [Foundational] • Target: The Logistic Differential Equation

For the logistic model $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$, at what population size is the growth rate $\frac{dP}{dt}$ maximized?

a. 250

b. 500

c. 125

d. 375

26 [Foundational] • Target: The Logistic Differential Equation

What is $\lim_{t \to \infty} P(t)$ for a logistic population with carrying capacity 800 if $P(0) = 50$?

a. 800

b. ∞

c. 400

d. 50

27 [Foundational] • Target: Analytical Solutions to Logistic Models

What is the general form of the solution to the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$?

a. $P(t) = \frac{M}{1 + Ae^{-kt}}$

b. $P(t) = Me^{kt}$

c. $P(t) = M - Ae^{-kt}$

d. $P(t) = \frac{M}{1 - Ae^{kt}}$

28 [Foundational] • Target: Analytical Solutions to Logistic Models

If $P(t) = \frac{1000}{1 + 9e^{-0.2t}}$, what is the initial population $P(0)$?

a. 100

b. 1000

c. 90

d. 10

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Analytical Solutions to Logistic Models**

What is the concavity of the logistic curve when $P(t) > \frac{M}{2}$?

a. Concave down ($P''(t) < 0$)

b. Concave up ($P''(t) > 0$)

c. Zero concavity

d. Undefined

30 **[Intermediate]** • **Target: Advanced Slope Fields with Autonomous Equations**

In the autonomous differential equation $\frac{dy}{dx} = y(y-3)(y+2)$, what are all equilibrium solutions?

a. $y = 0, y = 3, y = -2$

b. $y = 3, y = -2$ only

c. $y = 0$ only

d. $x = 0, x = 3, x = -2$

31 **[Intermediate]** • **Target: Advanced Slope Fields with Autonomous Equations**

For $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, which equilibrium solution is stable (an attractor)?

a. $y = 4$

b. $y = 0$

c. Both are stable

d. Neither is stable

32 **[Intermediate]** • **Target: Advanced Slope Fields with Autonomous Equations**

If $y(0) = 6$ in $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, what is $\lim_{x \to \infty} y(x)$?

a. 4

b. 6

c. ∞

d. 0

33 **[Intermediate]** • **Target: Euler's Method Iterations**

A population of bacteria is modeled by the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{L})$, where $P(t)$ is the population at time t in hours, k is the growth rate constant, and L is the carrying capacity. It is observed that $\frac{dP}{dt} = 200$ when $P = 100$, and the maximum growth rate occurs when $P = 500$. Find the carrying capacity L of this population.

a. 1000

b. 750

c. 1200

d. 500

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: The Logistic Differential Equation

Consider the logistic differential equation $\frac{dP}{dt} = kP(M - P)$, where $P(t)$ is the population at time t, k is the growth rate constant, and M is the carrying capacity. If $\frac{dP}{dt} = 0.05P(1000 - P)$, what is the carrying capacity of the population?

a. 1000

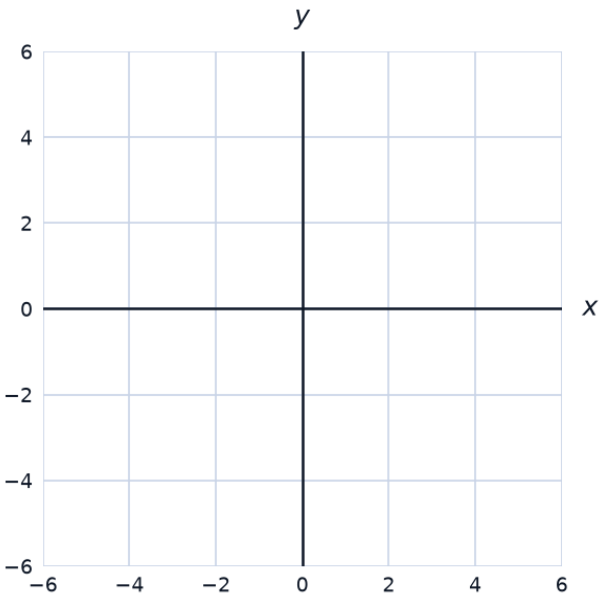
b. 0.05

c. 50

d. 0.00005

35 [Intermediate] • Target: Analytical Solutions to Logistic Models

Logistic growth model is $\frac{dP}{dt} =$



a. kP

b. $kP(1 - P/M)$

c. $k(M - P)$

d. P^2

36 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

A colony grows according to $\frac{dP}{dt} = 0.11P\left(1 - \frac{P}{1100}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=1100,\ P=550

b. M=1100,\ P=1100

c. M=550,\ P=1100

d. M=2200,\ P=550

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Euler's Method Iterations

A colony grows according to $\frac{dP}{dt} = 0.44P\left(1 - \frac{P}{440}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=880,\ P=220

b. M=440,\ P=220

c. M=440,\ P=440

d. M=220,\ P=440

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Logistic Differential Equation • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A colony grows according to $\frac{dP}{dt} = 0.66P\left(1 - \frac{P}{660}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=660,\ P=660

b. M=660,\ P=330

c. M=1320,\ P=330

d. M=330,\ P=660

39 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 1.01 Integration by Parts

A colony grows according to $\frac{dP}{dt} = 0.88P\left(1 - \frac{P}{880}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=880,\ P=440

b. M=440,\ P=880

c. M=1760,\ P=440

d. M=880,\ P=880

40 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A colony grows according to $\frac{dP}{dt} = 0.33P\left(1 - \frac{P}{330}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=660,\ P=165

b. M=330,\ P=330

c. M=165,\ P=330

d. M=330,\ P=165

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Euler's Method Iterations • Cross-Lesson: 3.01 Parametric Equations

A colony grows according to $\frac{dP}{dt} = 0.22P\left(1 - \frac{P}{220}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=220,\ P=220

b. M=220,\ P=110

c. M=440,\ P=110

d. M=110,\ P=220

42 [Challenging] • Target: The Logistic Differential Equation • Cross-Lesson: 3.02 Vector-Valued Functions

For $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{2000}\right)$, find the instantaneous growth rate when $P = 1000$.

a. -25

b. 25.5

c. 25

d. 50

43 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $\frac{dP}{dt} = 0.25P\left(1 - \frac{P}{800}\right)$, find the instantaneous growth rate when $P = 300$.

a. 93.75

b. -46.875

c. 46.875

d. 75

44 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 4.02 Area in Polar Coordinates

For $\frac{dP}{dt} = 0.55P\left(1 - \frac{P}{350}\right)$, find the instantaneous growth rate when $P = 140$.

a. 46.2

b. -46.2

c. 92.4

d. 77

45 [Challenging] • Target: Euler's Method Iterations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $\frac{dP}{dt} = 0.85P\left(1 - \frac{P}{60}\right)$, find the instantaneous growth rate when $P = 20$.

a. 11.3333

b. 17

c. 22.6667

d. -11.3333

© ClipSAT — clipsat.org | All Rights Reserved